- ★ A tangent line to a circle can be analysed with pre calculus mathematics. To analyse a line tangential to a general graph, you need calculus.
- ★ The area of a rectangle can be analysed with pre calculus mathematics. To analyse the area under a general curve, you need calculus.

Each of these situations involves the same general strategy, the reformulation of pre calculus mathematics through the use of a limit process. So, one way to answer the question 'What is calculus?' is to say that calculus is a 'limit machine' that involves three stages. The first stage is pre calculus mathematics such as the slope of a line or the area of a rectangle. The second stage is the limit process and the third stage is a new calculus formulation, such as a derivative or integral.

Pre calculus Mathematics	⇒	Limit Process	⇒	Calculus

It is cautioned that those who try to learn calculus as if it were simply a collection of new formulae rather than as a process, will miss a great deal of understanding, self-confidence and satisfaction.

Learning Objectives

On completion of this chapter, the students are expected to

- visualize the concept of limit / continuity through geometric process.
- relate the concept of limit / continuity with every day life activities.
- assimilate limit / continuity as the heart and spirit of calculus.
- understand limit / continuity as an operation (operator) to measure / quantify / mathmatize changes in physical world.
- concretize the concept of limit / continuity via illustrations of real life related situations.

9.2 Limits

9.2.1 The calculation of limits

The notion of a limit, which we will discuss extensively in this chapter, plays a central role in calculus and in much of modern mathematics. However, although mathematics dates back over three thousand years, limits were not really understood until the monumental work of the great French mathematician Augustin – Louis Cauchy and Karl Weierstrass in the nineteenth century, the age of rigour in mathematics.

In this section we define limit and show how limits can be calculated.

Illustration 9.1

We begin by looking at the function $y = f(x) = x^2 + 3$. Note that f is a function from $\square \to \square$.

Let us investigate the behaviour of this function near $x = 2$. We can use two sets of x values : one set that approaches 2 from the left (values less than 2) and one set that approaches 2 from the right (values greater than 2) as shown in the table.

	x approaches 2 from the left							*x* approaches 2 from the right					
x	1.7	1.9	1.95	1.99	1.999	1.9999	2	2.0001	2.001	2.01	2.05	2.1	2.3
f(x)	5.89	6.61	6.8025	6.9601	6.99601	6.99960001	7	7.0040001	7.004001	7.0401	7.2025	7.41	8.29

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